1 Background and Notations

1.1 Poisson Processes

Definition 1. A counting process $N = (N(t))_{t\geq 0}$ is a sequence of non-negative random variables such that :

— $N(0) = 0$,

— $N(t)$ takes values in the set of non-negative integers,

— $N(t)$ is non-decreasing and right-continuous.

Definition 2. A Poisson process with rate $\lambda > 0$ is a counting process $N = (N(t))_{t\geq 0}$ such that :

— $N(0) = 0$,

— The process has independent increments,

— The number of events in any interval of length t follows a Poisson distribution with parameter λt.

1.2 Conditional Poisson Processes

Definition 3. A conditional Poisson process is a counting process $N = (N(t))_{t\geq 0}$ such that, given a stochastic intensity function $\lambda(t)$, the process behaves like a Poisson process with rate $\lambda(t)$ at time t.

1.3 Cramér-Lundberg Model

We formally define the surplus process as follows :

$$R(t) = u + ct - S(t), \tag{1}$$

with u the initial reserve, c the premium rate, and $S(t)$ the aggregate claims up to time t.

In order to model $S(t)$, we consider the classical Cramér-Lundberg model which assumes that :

$$S(t) = \sum_{i=1}^{N(t)} X_i, \tag{2}$$

where $N(t)$ is a Poisson process with rate $\lambda > 0$, representing the number of claims up to time t. The X_i are i.i.d. random variables representing the claim sizes, **independent** of $N(t)$, with common distribution F_X.